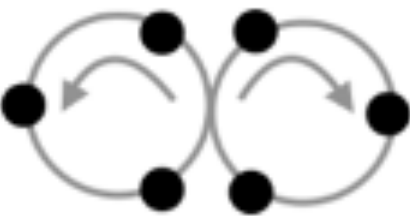


2pi/3
(or better, 2π/3). Maybe it's cooler than the Turbo!



Read this *two pi on three*. This one is a late bloomer, considering how simple it is and how long people have been juggling and moving in groups of three.

Beat, base
count (½ cycle
in table)

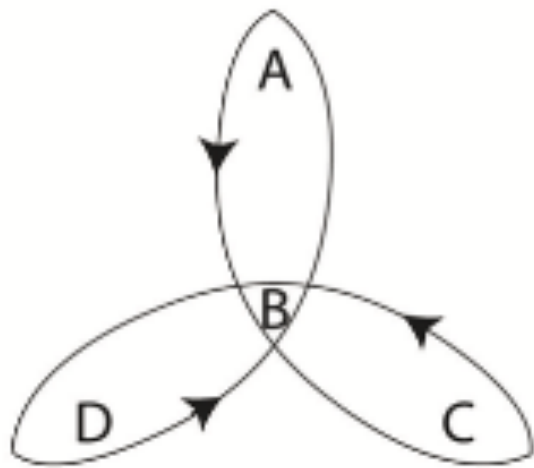
	1	2	3		1	2	3
				A	B	C	
				B	A		C
				C		A	B

Three jugglers, **A**, **B**, and **C**, all move through the figure eight over the course of six beats on the base count. In contrast to a *Turbo*, there is not a feeder being swapped here; all folks are moving all the time. Relative to the figure in top right, there are six positions, and at any point jugglers occupy three alternating positions. The rule is pass and move to the next position. Any count works; 5-count is pleasant, 4-count is fun, 3-count is more fun! Recent variations alternate the count, such as 1-4 on a 7 for a 3-count/4-count alternation. In any count, it's **pass-one, pass-the other, self (i.e. phantom)**. Whatever is the base count, the pattern has a gap where you can plug in any other juggler or pattern that matches that count (including three other jugglers doing a 2π/3). Looks like two pies when viewed from above, and it's 3 jugglers. Plus, each circle in the layout is divided into 3, hence the name. Prof Mike invented 2π/3 in summer 2017, but the popular version is Mel's variation (shown above), in which the first recipient in the pair moves through. In the original, the first recipient goes backwards and the non-recipient goes through (equivalently, start the passing sequence with **A-C**. This is a curiously different and more difficult version owing to backwards moves out of the middle..

Three Leaf Clover (Standard and Run-Over)

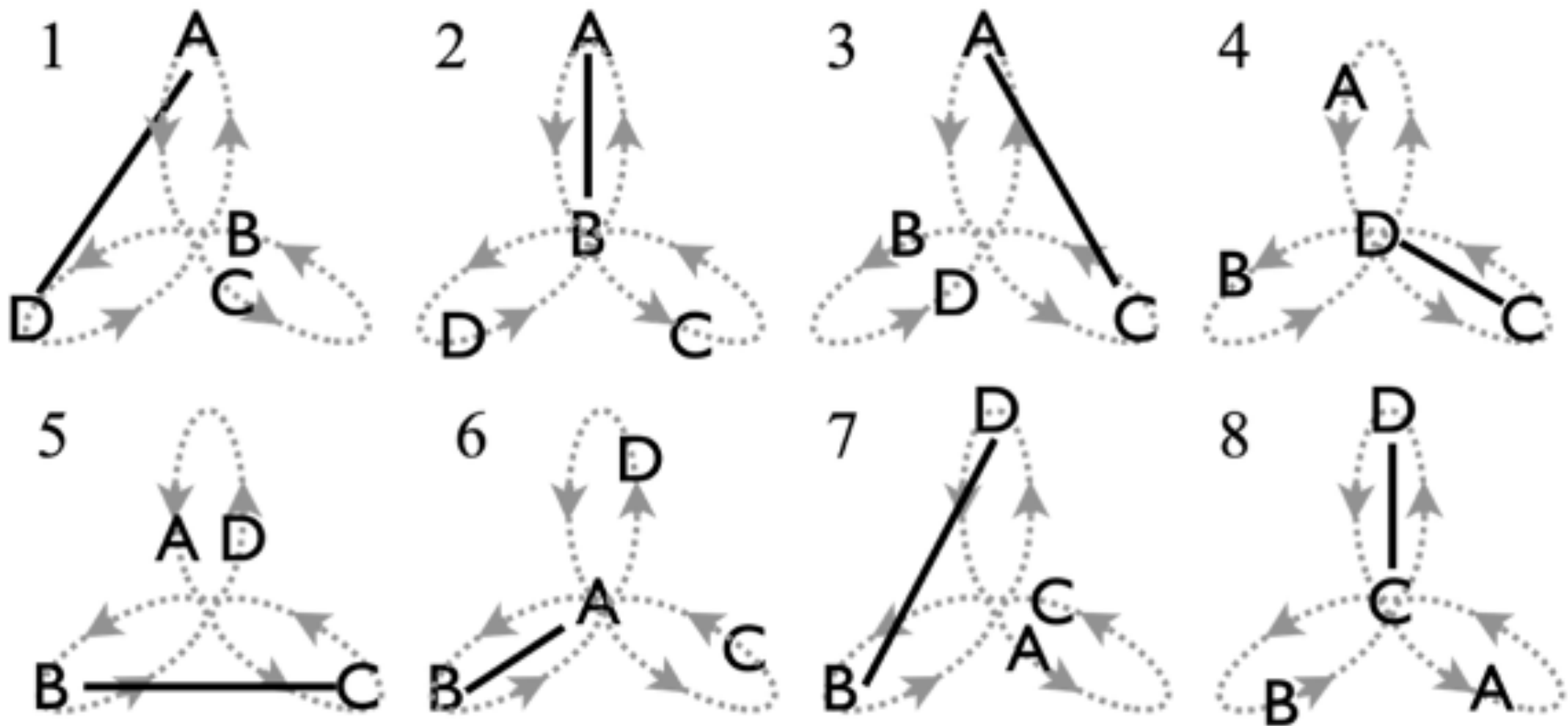


Set up 1 per leaf and one middle.



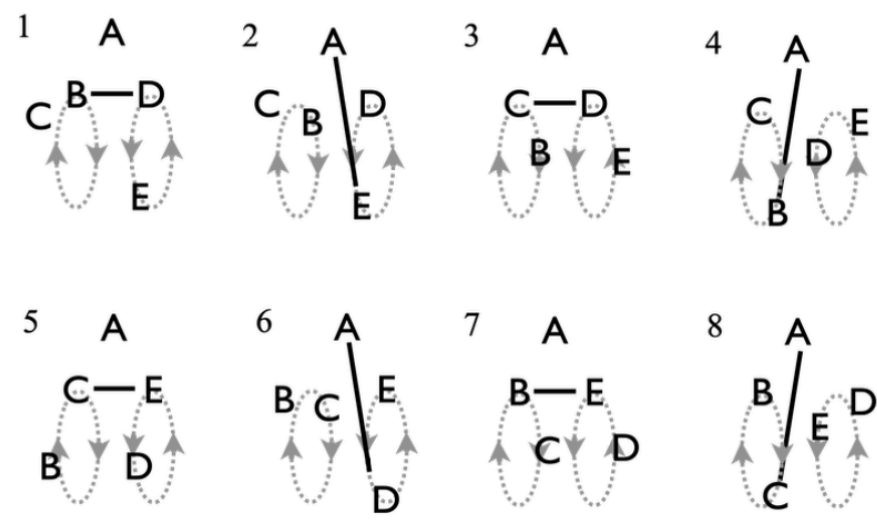
	Beat, base (e.g. 2 count)							
Juggler	1	2	3	4	5	6	7	8
A	D	B	C	B		B		B
B		A		A	C	A	D	A
C		D	A	D	B	D		D
D	A	C		C		C	B	C

Rule: Feeder sweeps right to left, then **2-in-2-out** move along arc to replace the person on your left.
Pro tip: After your feed, move out to about 2 o'clock to your right, and also rotate clockwise (from above), opening up to your middle pass as you translate left.
Variations: The standard version is a 2-count base beat, but 3-count and 4-count are fun too. Over the last year or so we often do this on one of the variable counts, like 1-3-on-a-5-count, or 1-3-5-on-a-7-count. Additional (grey) passes are optional; we call that a *Run-Over-Clover*; it makes the move **1-in-1-out**, which goes nicely on a 3-count base rather than the faster 2-count. Watch the 3-count *run-over-clover* with colors, recorded during the COVID pandemic:
<https://www.youtube.com/watch?v=bJLnODIIHMI>



Local cycle: **A** starts the feed and ends one leaf to the left ready to start a second feed.

Eggbeater (original)



Feeder **A** - the *handle* - passes to 4 movers in 2 pairs of 2 spinning around each other. **B** and **C** form one *beater*, and **D** and **E** another. It's pretty good for beginner movers on the original base count (2-count), but you have to think because your count changes as you go: 6-count; 4-count; 6-count. Performed by Mad-5 at 1997 Madfest.

Beat (2-count base)									3-4-5 Eggbeater
	1	2	3	4	5	6	7	8	A beautiful variation emerges if A speeds up to 3-count from 4-count. Then movers have a 3-count gap between their stationary passes, a 4-count gap to move to the bottom of the bowl, and a 5-count gap to come back to the top. It's fast and uses both hands!
A		E		B		D		C	
B	D		E	A	D		E		
C	E		D		E		D	A	
D	B		C		B	A	C		
E	C	A	B		C		B		
The standard 2-count base, 8-beat cycle eggbeater is written on a 1-count base in the table below (row 2). The third shows how the 3-4-5 eggbeater works out on that 1-count base. In other words, it's the same table as left except on a 3/2-count base!									

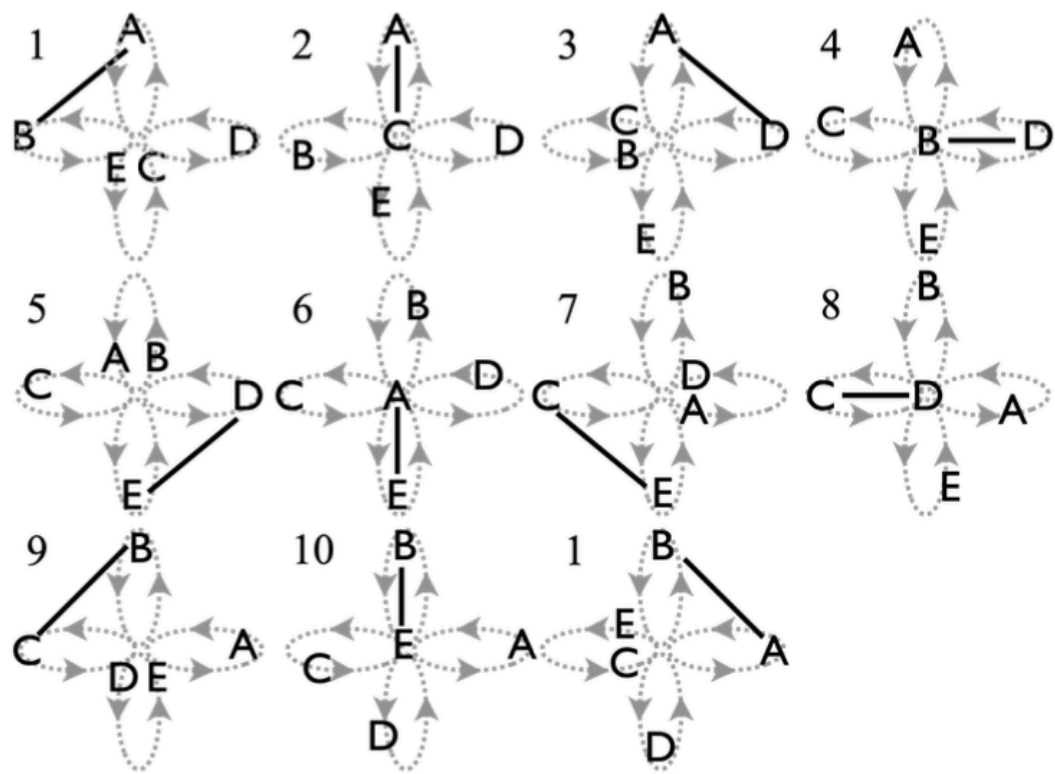
Light grey are extra passes in a Martin Frost variation born at 1996 Madfest. 3-4-5 was born about 2016. Mad-5 performed the original in the 90's. Mike proposed the handle-in-the batter, and possibly the 3-4-5.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		2		3		4		5		6		7		8	
1	2		3	4		5	6		7	8					

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	4	5	6	7	8								
1	2	3	4	5	6	7	8								

Four Leaf Clover

The deal with clover patterns is that the number of jugglers is one more than the number of leaves; there's always one juggler on each leaf and one extra in the middle, about to swap places with one of the leaves. This is a very pleasant 5 person moving pattern on a 2-count or 3-count base; all jugglers do the same thing in the steady state: feed three (right to left); 2-selves to move into middle; pass in middle; 4 selves to get out. That's a **2-in-4-out**, in contrast to the **2-in-2-out** three-leaf clover for four jugglers.



Shown left are 10 beats (local cycle) of 40 required to get everyone situated at the same place. The 11th beat has the whole pattern rotated a ¼ turn (technically, the passing sequence has completed a full cycle, even though the jugglers are repositioned!).

After feeding, you have 2 beats to move straight across, then 4 easy beats to move out left and set up for the next cycle. It can clog up in the middle if you're not careful with those 2 entering beats.

Patterns by MAJ.

	Beat (base count; e.g. 2 count)									
	1	2	3	4	5	6	7	8	9	10
A	B	C	D	C		E		E		
B	A	E		D		D			C	E
C		A		A			E	D	A	D
D			A	B	E	B		C		C
E		B			D	A	C	A		B

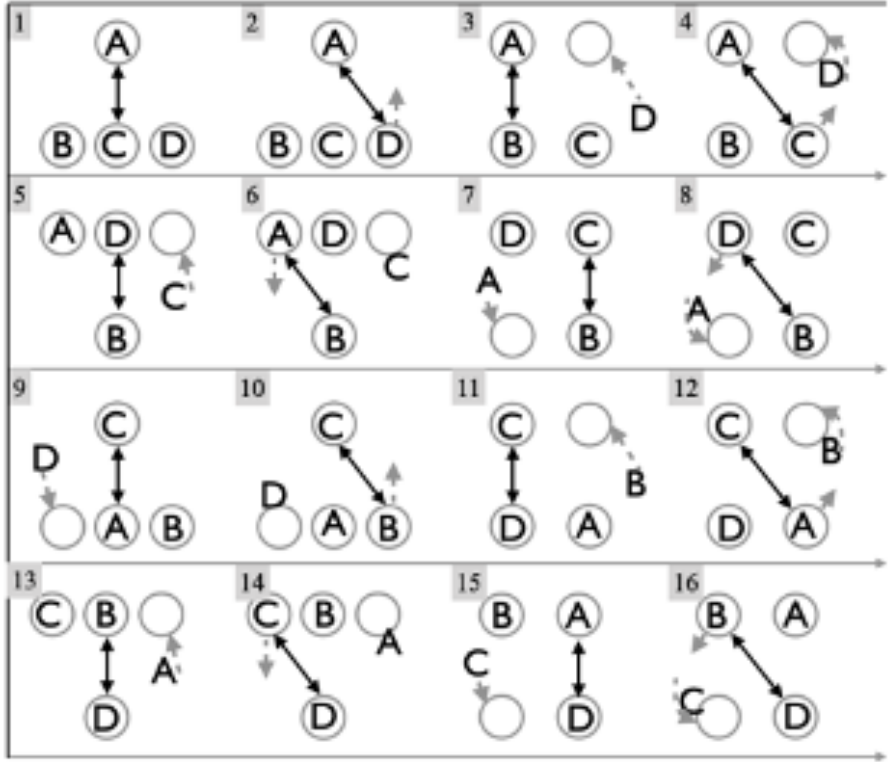
Variations: Aside from count variations (2 and 3 most common), *extra passes* are available to jugglers moving through (one version is noted in grey in table left).

Watch the 4-leaf clover on 3-count (<https://www.youtube.com/watch?v=ikgU-CTVB9o>)

El Niño

El Niño is a rotating 1-count feed and one of the crown jewels of the MAJ.

	Beat, 1-count base, left-hand start, grey=feeder, black=mover																<i>El Niño</i> dates to August, 1997, by Peter Kaseman and MAJ
Juggler	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
A	C	D	B	C		B	⇒		C			C	⇒		D		
B	⇒		A		D	A	C	D		C	⇒		D			D	
C	A			A	⇒	B		A	B	D	A		D	⇒			
D		A	⇒	B			B	⇒		C		B	C	A	B		



Start **left-handed** (*the modern start!*). Notice there's a single switch-side move connecting a typewriter-1-count feed of 3-counters and a 2-on-2 double three count box. Initially, the version here was considered the "fast version", with slower versions stepping between the feed and box. The version reported here smooths all this into a beautiful 16 beat moving pattern. **Rule:** Move forward off the feeder's left side of the feed, while passing a right and similarly receiving a pass into your inside (left) hand.

This challenging pattern is quite doable once passers master the stationary 1-count feed and the 2 on 2 double 3-count box. It is essentially a continuous switch-sides 1-count feed. A fun 5-person practice has 4 doing *El Niño* and 1 sub who tags in for the first person to drop!